

McClaw: a Dual-Hand Motion-Controlled Mobile Robotic Arm

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I. OBJECTIVES

General Objective

To design, develop, and implement the *McClaw* dual-hand motion-controlled mobile robotic arm system using motion sensors (gyroscope and accelerometer), a flex sensor, and wireless communication for real-time robotic navigation and manipulation.

Specific Objectives

- To implement a left-hand gesture control system using a motion sensor (gyroscope and accelerometer) with an Arduino Nano transmitter for real-time navigation of the mobile robot (forward, backward, left, right, and stop).
- To implement a right-hand gesture control system using a motion sensor (gyroscope and accelerometer) and a flex sensor with an Arduino Nano for controlling robotic arm movement and claw gripping actions.
- To establish a stable wireless communication system between the dual-hand controllers and the robotic platform.
- To integrate a motor-driven mobile base using DC motors and a motor driver (L298N or equivalent) for directional movement.
- To develop and implement a multi-axis robotic arm system using servo motors for precise manipulation and object handling.
- To design and apply a control algorithm that translates dual-hand gestures into simultaneous and coordinated robot navigation and arm operations.
- To evaluate the system's performance in terms of responsiveness, stability, and usability during navigation and pick-and-place tasks.

II. SCOPE AND LIMITATIONS

Scope

The system implements a dual-hand gesture control interface for a mobile robotic arm system.

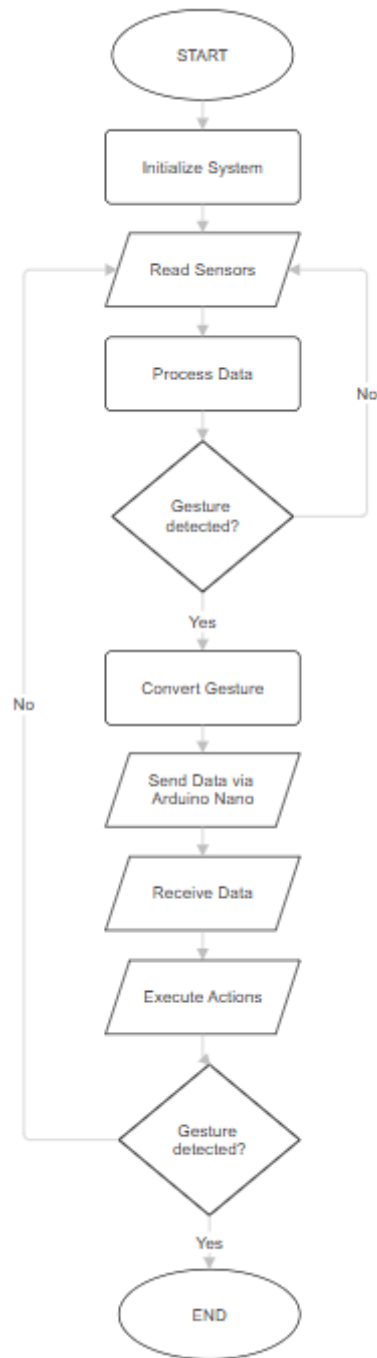
- The left-hand controller uses a motion sensor (gyroscope and accelerometer – MPU6050) for robot navigation (forward, backward, left, right, stop).
- The right-hand controller uses a motion sensor (gyroscope and accelerometer – MPU6050) and a flex sensor for robotic arm movement and claw control.

- The system uses Arduino Nano microcontrollers with NRF24L01 2.4GHz transceiver modules for wireless communication between controllers and the robot.
- The mobile robot is built using a DC motor-driven wheeled base for movement.
- A robotic arm with servo motors is used for object manipulation and gripping.
- The system operates in real-time and is intended for indoor or laboratory-scale environments.

Limitations

- The system depends on stable hand gestures; sudden or unintended movements may affect accuracy.
- The motion sensor (MPU6050) is angle-dependent and may not provide high precision in all orientations.
- The flex sensor has limited sensitivity and is mainly used for simple open/close claw actions.
- Wireless communication using NRF24L01 modules may experience minor delays or packet loss depending on interference and distance.
- The system is limited to short-range operation due to NRF24L01 communication constraints.
- Dual-hand control may require practice for smooth coordination.
- The system is not suitable for heavy-duty or industrial-grade robotic applications.

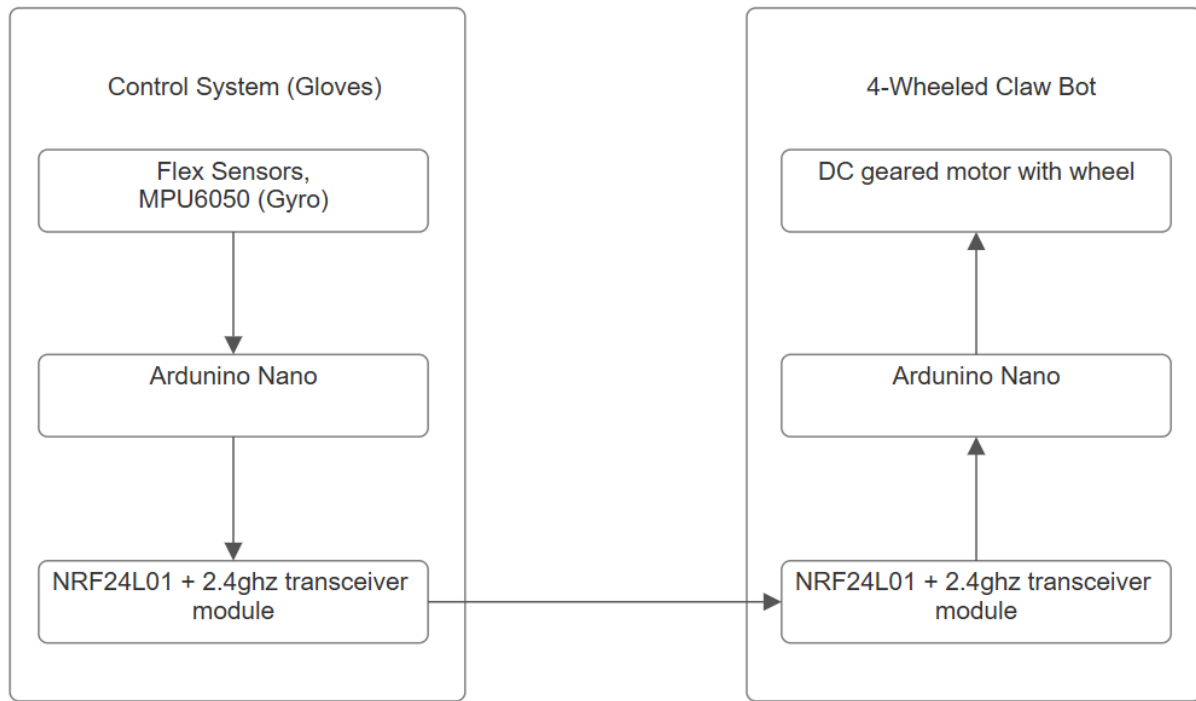
III. FLOWCHART



The operational logic of the system is outlined in the flowchart above, beginning with system initialization and a continuous sensor polling loop. The process starts by reading raw sensor data and processing it to determine if a specific movement has occurred; if no gesture is recognized, the system loops back to the reading stage. Once a gesture is detected, the data is converted into a command and transmitted wirelessly via the Arduino Nano. The receiver then captures this data and executes the corresponding physical action on the robot. Finally, the system checks for a termination gesture to either end the process or return to the initial sensor-reading phase for continued operation.

IV. HARDWARE DEVELOPMENT

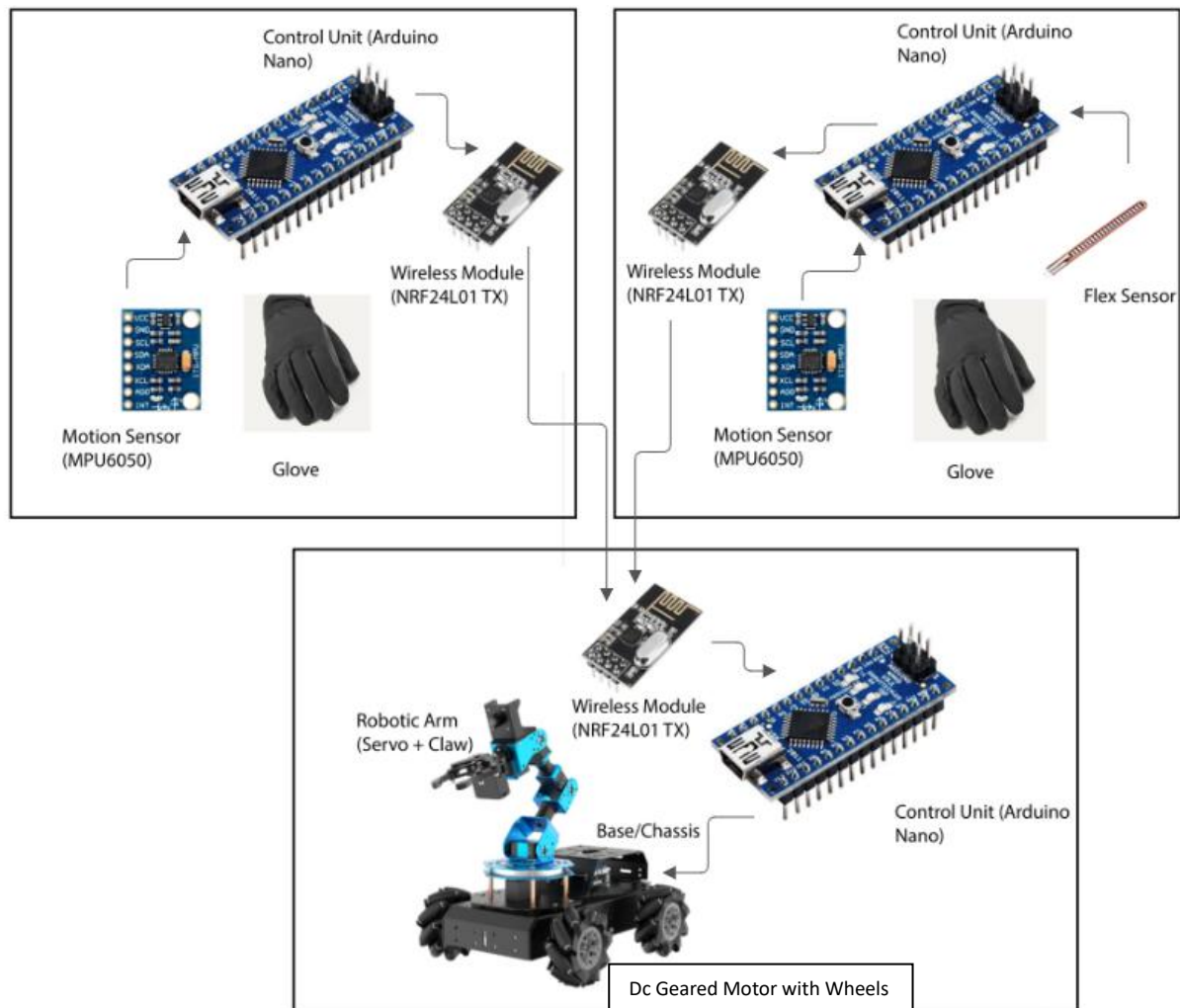
A. Block Diagram



The system operates as a wireless feedback loop that translates human hand gestures into robotic movement. As illustrated in the figure above, the process begins at the Control System (Glove), where Flex Sensors and an MPU6050 gyroscope detect the bending of fingers and the orientation of the hand. This motion data is processed by an Arduino Nano, which converts the physical movements into digital commands and sends them to an nRF24L01 transceiver for wireless broadcast. On the receiving end, the 4-Wheeled Claw Bot uses a matching transceiver to capture these signals and relay them to a second Arduino Nano. This receiver chip interprets the instructions to drive DC geared motors, allowing the robot to drive, turn, or operate its claw in real-time synchronization with the user's hand gestures.

B. Pictorial Diagram

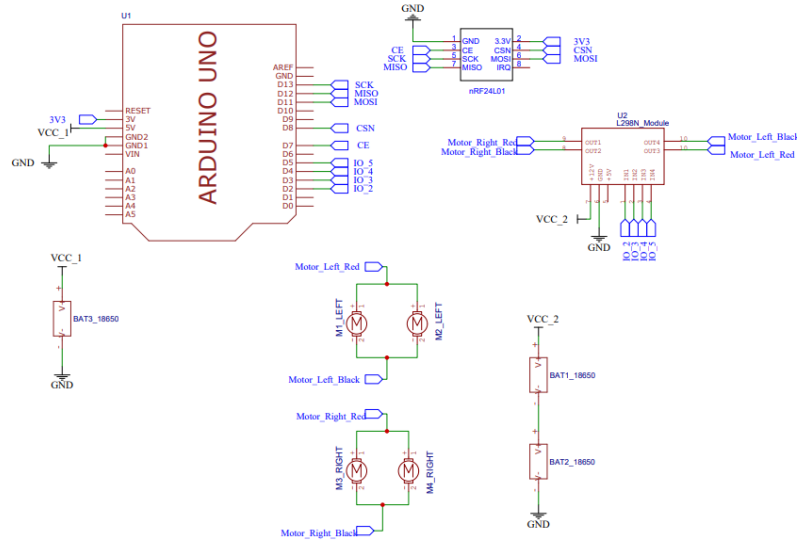
The pictorial diagram shows the overall layout of the system by presenting the physical arrangement of the glove controllers and robot base. It illustrates how the main components are grouped and connected through wireless communication.



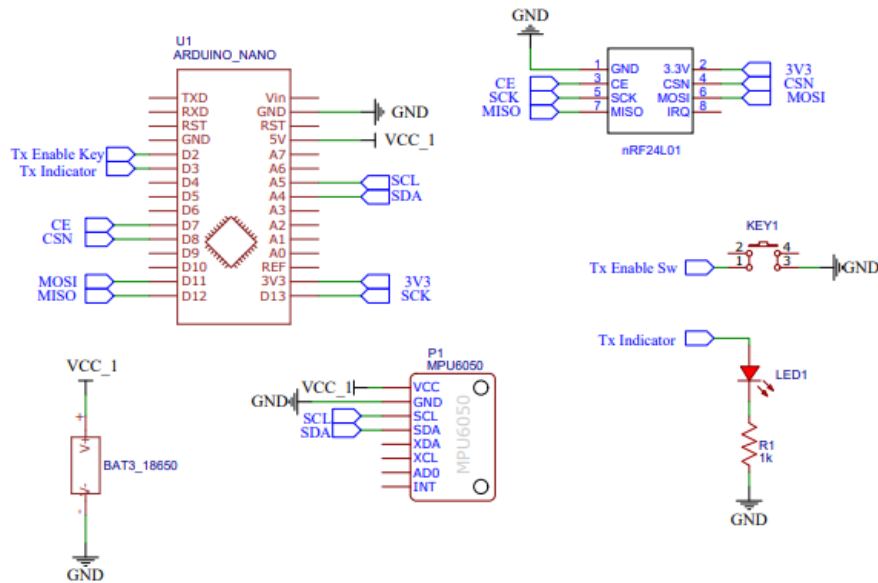
The system employs a dual-transmitter architecture to enable bi-manual wireless control of a robotic platform. As illustrated in the figure above, two independent glove units each containing an Arduino Nano, MPU6050 motion sensor, and nRF24L01 module capture hand orientation and finger flex data. These control units broadcast data packets to a central receiver on the robot, where an onboard Arduino Nano decodes the telemetry to drive an omnidirectional chassis and a multi-axis robotic arm. This configuration allows for the real-time translation of complex human gestures into synchronized locomotion and claw manipulation.

C. Schematic Diagram

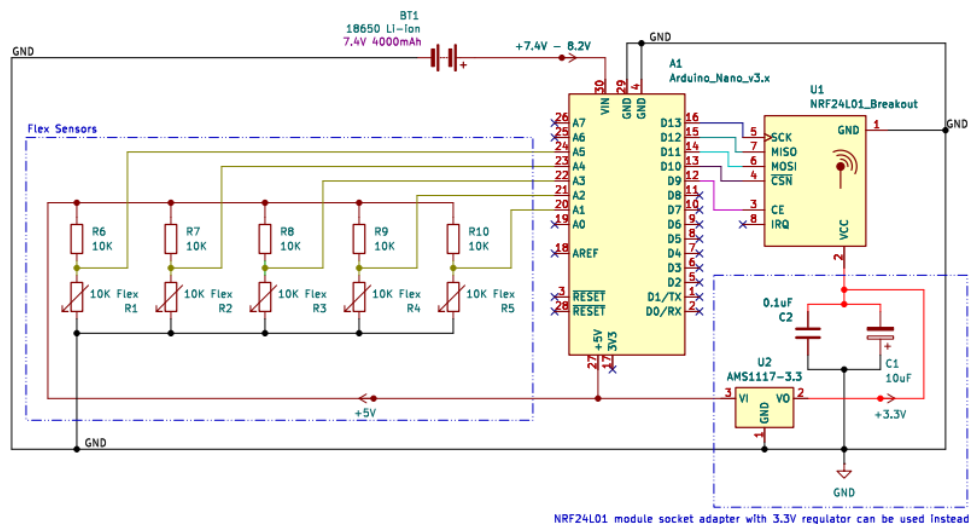
The following technical overview outlines the hardware architecture of the gesture-controlled robotic system. By integrating motion-sensing electronics with wireless communication protocols, the system facilitates an intuitive link between human kinetic input and mechanical execution.



The schematic illustrated above details the electrical configuration of the Receiver (4-Wheeled Bot) unit, which serves as the physical actuator for the system. An Arduino Uno acts as the central processor for basic controls, utilizing an nRF24L01 transceiver to capture wireless data packets transmitted by the control glove. To handle the high current requirements of the propulsion system, the microcontroller interfaces with an L298N motor driver module, which regulates power from a dedicated multi-cell 18650 battery supply. This driver manages four DC geared motors wired in parallel pairs for the left and right sides enabling the robot to execute directional movement and differential steering based on the received gesture commands.



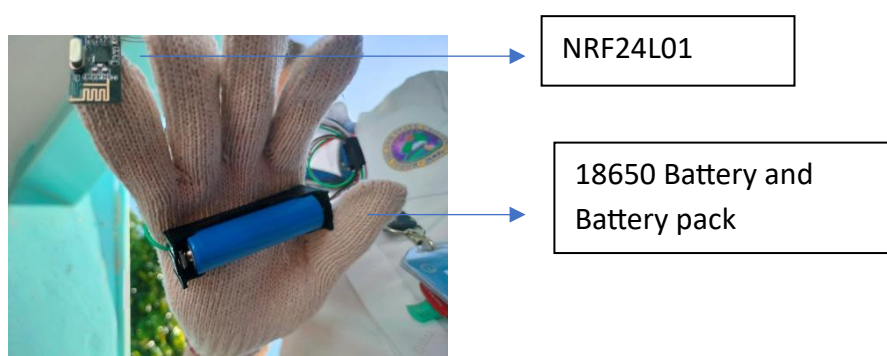
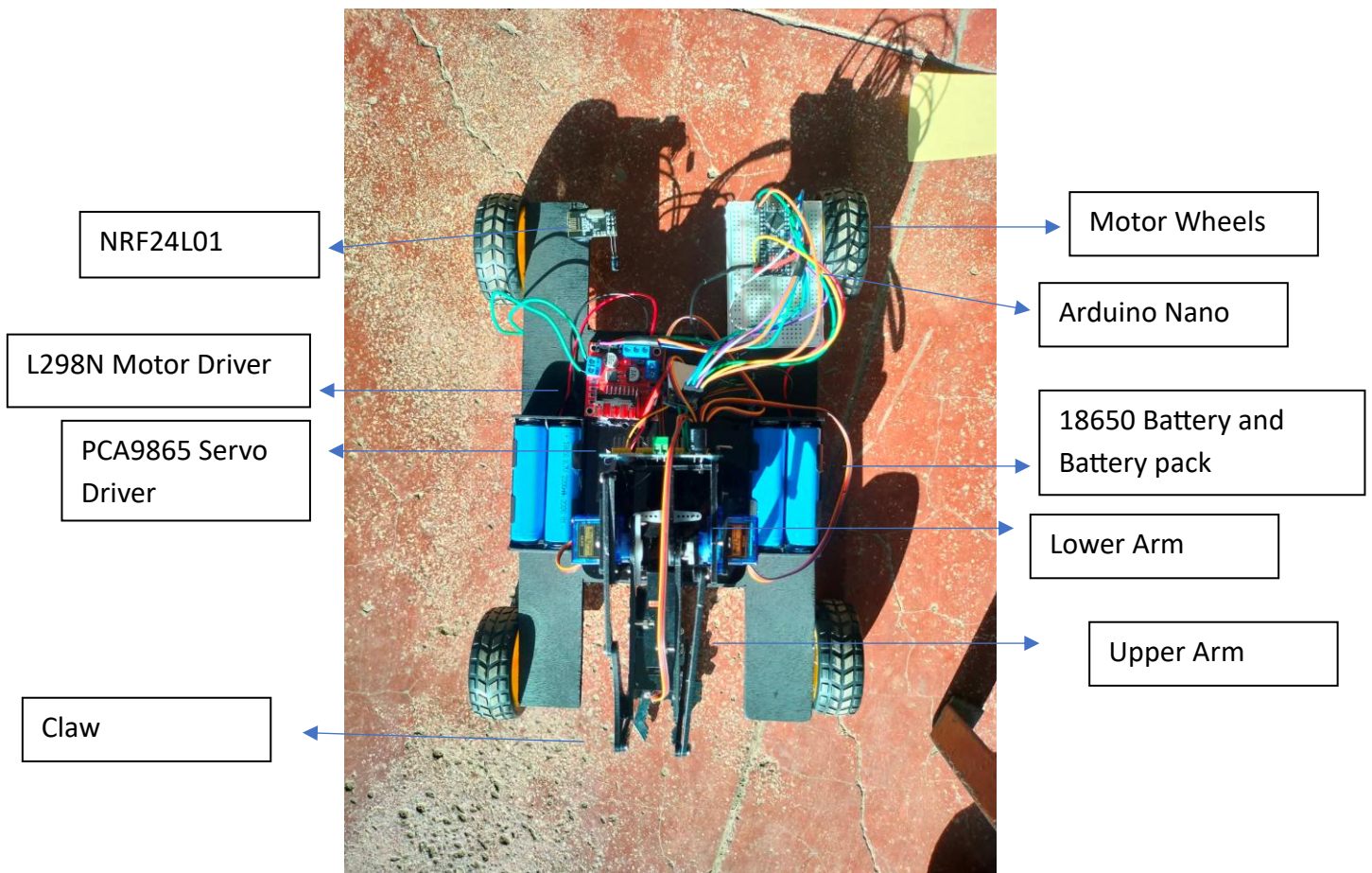
The schematic illustrated above depicts the electronic framework of the control interface, centered around an Arduino Nano microcontroller. The system is powered by an 18650 battery and utilizes an MPU6050 sensor to capture hand orientation data via Inter-Integrated Circuit communication. A tactile "Tx Enable" switch allows the user to engage the transmission, while an LED indicator provides visual confirmation of the broadcast status. These processed signals are then sent to the nRF24L01 transceiver module through SPI pins (MOSI, MISO, SCK) to be beamed wirelessly to the robotic platform

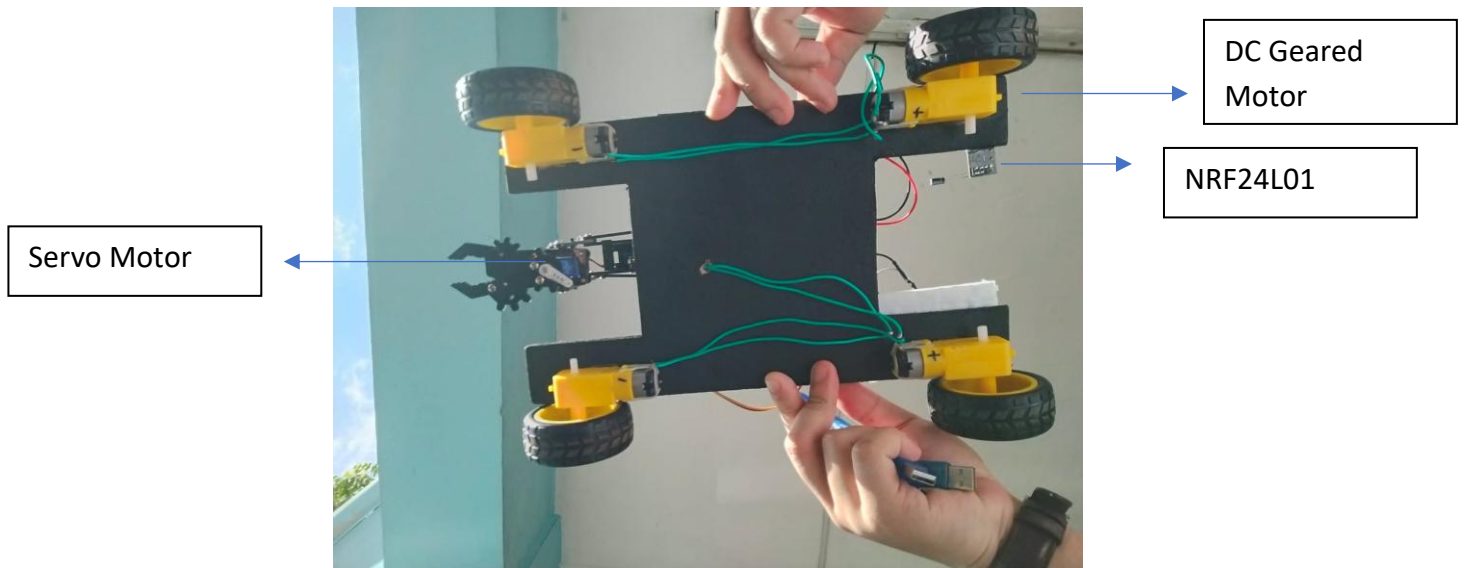


The schematic illustrated above depicts the sensor interface and power management circuitry for the transmitter unit. The design incorporates flex sensors arranged in a voltage divider configuration with 10k ohm fixed resistors, allowing the Arduino Nano to translate physical displacement into measurable analog signals. Power is provided by a 7.4V Li-ion battery, which is stepped down to a stable 3.3V via an AMS1117-3.3 voltage regulator to ensure

consistent performance for the nRF24L01 transceiver. The resulting gesture data is processed and broadcast wirelessly via the SPI communication bus, forming the primary control link for the robotic platform's movement and actuation.

V. PHYSICAL ARCHITECTURE AND ENCLOSURE - figure (design and component layout of the system)





VI. IMPLEMENTED DESIGN - figure/image of the actual -system





The system functions as a wireless teleoperation link between a hand-mounted controller and a mobile robot. In the image, the glove acts as the input device; it contains an inertial measurement unit (IMU) that tracks the orientation of the hand in real-time. When the user tilts their hand forward, backward, or side-to-side, the onboard microcontroller processes these orientation changes into specific directional data. This data is then broadcasted through a wireless transmitter module, allowing the hand's tilt to serve as a virtual joystick for the vehicle.

The four-wheeled robot serves as the receiver and actuator. It is equipped with a wireless receiver that catches the incoming signals from the glove. The central microcontroller on the chassis interprets these signals and sends electrical pulses to a motor driver, which manages the high-current power from the blue lithium-ion batteries to the four DC motors. For instance, a "tilt forward" signal from the glove tells the motor driver to spin all wheels forward, while a "tilt left" signal might slow down the left wheels and speed up the right ones to execute a turn. This continuous loop of sensing, transmitting, and motor execution allows the robot to mirror the physical movements of the user's hand instantly.

VII. CODE SNIPPETS

A. MPU6050 Initialization (Sensor Setup)

Code:

```
Wire.begin();
Wire.beginTransmission(MPU_addr);
Wire.write(0x6B);
Wire.write(0);
Wire.endTransmission(true);
```

Explanation: This code initializes the MPU6050 motion sensor by starting I2C communication and resetting the sensor to activate its accelerometer and gyroscope functions.

B. Reading Motion Sensor Data

```
Code:
Wire.beginTransmission(MPU_addr);
Wire.write(0x3B);
Wire.endTransmission(false);
Wire.requestFrom(MPU_addr, 14, true);
axis_X = Wire.read() << 8 | Wire.read();
axis_Y = Wire.read() << 8 | Wire.read();
axis_Z = Wire.read() << 8 | Wire.read();
```

Explanation: This reads raw acceleration and gyroscope data from the MPU6050 sensor, which is used to detect hand movement direction and orientation.

C. Flex Sensor Gripper Control

```
Code:
int flex_sensorip = analogRead(A0);
if (flex_sensorip > 750)
  gripper = 0;
else
  gripper = 180;
servo_3.write(gripper);
```

Explanation: This code reads the flex sensor value and controls the robotic gripper. When the finger bends, the gripper closes; otherwise, it opens.

D. Arm Movement Control (Servo Mapping)

```
Code:
int xAng = map(axis_X, minVal, maxVal, -90, 90);
if (x >= 0 && x <= 60)
```

```
{int mov1 = map(x, 0, 60, 0, 90);  
servo_1.write(mov1);}
```

Explanation: This converts motion sensor values into servo motor angles, allowing the robotic arm to move based on hand tilt direction.

E. Left/Right Movement Control

Code:

```
if (y >= 0 && y <= 60)  
{int mov3 = map(y, 0, 60, 90, 180);  
servo_4.write(mov3);}
```

Explanation: This controls left and right movement of the robotic arm by mapping hand tilt on the Y-axis to servo rotation angles.

VIII. BILL OF MATERIALS

A. Materials Table

This section lists all the materials and components used in the system, including their quantities, unit prices, and total costs. These materials are used for the glove-based controller and the robotic base.

Item	Description	Qty	Unit Cost (₱)	Total Cost (₱)
Arduino Nano	Microcontroller	3	₱299	₱897
MPU6050	Motion sensor	2	₱115	₱230
NRF24L01 Module	2.4GHz transceiver	3	₱69	₱207
Flex Sensor	Bend sensor	1	₱809	₱809
DC Geared Motors	Motor With wheels	2	₱65	₱130
Robot Chassis	H-type frame	1	₱237	₱237

Rechargeable Battery	3.7–5V	2	₱49	₱98
Battery Pack	7V supply	1	₱76	₱76
Jumper Wires	Connection wires	1 set	₱43	₱43
PCB	Circuit board	1	₱61	₱61
Glove	Wearable base	2	₱20	₱40
AMS1117-3.3	Power Supply Module	5	₱12	₱60

Total Value: ₱ 2,888

IX. CONCLUSION AND RECOMMENDATION

The McClaw dual-hand motion-controlled robotic system was successfully designed and developed using motion sensors, flex sensors, Arduino Nano microcontrollers, and NRF24L01 wireless communication modules. The system effectively translates hand gestures into real-time commands for both mobile robot navigation and robotic arm control. Through the integration of hardware components and control logic, the project demonstrates that gesture-based control can be an efficient and intuitive method for operating a robotic system in a laboratory-scale environment.

I would recommend further improving the system by enhancing the accuracy and stability of the motion sensors through better calibration and signal filtering techniques. I would also suggest upgrading the wireless communication system to reduce possible delays or interference during operation. In future development, I would consider using more precise sensors and adding feedback mechanisms such as vibration or visual indicators to improve user interaction. Additionally, I would recommend refining the ergonomic design of the glove to increase comfort and extending the capabilities of the robotic arm for more complex object manipulation tasks.